

ADITYA SINGH

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RESEARCH EXPERIENCE

- **Department of Chemistry, IIT Delhi**  June 2026 - Present
PhD Student - Under Prof. Dibyajyoti Ghosh New Delhi, India
 - **Research Area:** Electronic structure and excited-state phenomena of two-dimensional halide perovskites using first-principles calculations.
- **Department of Chemistry, IIT Delhi**  July 2025 - May 2026
Masters Thesis Student - Under Prof. Dibyajyoti Ghosh New Delhi, India
 - **Project Title:** Computational Insights into Hydrogen-Bond Driven PLQY Enhancement in Deep-Blue Copper-Iodide Hybrids.
- **Department of Chemistry, IIT Delhi**  November, 2024 - December, 2024
Winter Internship Project Student - Under Prof. Ravi P. Singh New Delhi, India
 - **Project :** Synthetic studies related to [1,2]-Phospha-Brook rearrangement
 - Assisted in substrate synthesis and reaction optimization for stereoselective reductive coupling reactions based on umpolung strategies.

EDUCATION

- **Indian Institute of Technology, Delhi** June, 2026 - Present
Doctor of Philosophy (PhD) in Chemistry New Delhi, India
 - Supervisor: Prof. Dibyajyoti Ghosh
- **Indian Institute of Technology, Delhi** July, 2024 - May, 2026
Master of Science in Chemistry New Delhi, India
 - CGPA: 9.1/10.00
- **University of Allahabad** November, 2021 - June, 2024
Bachelor of Science in Physics, Chemistry and Mathematics Prayagraj, India
 - Division: First Class
 - Top rank in Department of Chemistry
- **Government Inter College Prayagraj** July, 2019 - April, 2021
Class XII (Higher Secondary Certificate), UP Board Prayagraj, India
 - Percentage: 85.60%
 - Secured top rank in school; 99/100 in Mathematics, 92/100 in Physics.
- **Madan Mohan Malviya Inter College Prayagraj** July, 2017 - April, 2019
Class X (Secondary School Certificate), UP Board Prayagraj, India
 - Percentage: 92.33%
 - Ranked 8th in Prayagraj District in the Secondary School Examination, and was awarded by the Chief Minister of Uttar Pradesh.

SKILLS

- **Programming Languages:** Python, Bash Scripting
- **Operating Systems:** Linux, Windows, MacOS.
- **Libraries :** PyTorch, TensorFlow and Keras, ScikitLearn, Pandas, Numpy, Matplotlib, Seaborn
- **Document Creation:** Microsoft Office Suite, Overleaf (LaTeX Editor).
- **Modelling Software:** Gaussian, VMD, PyMOL, Avogadro, ChemDraw, VESTA, Olex
- **Packages:** VASP (quantum mechanical calculations, DFT).
- **Data Analysis & ML (Foundational) :** Familiarity with graph-based representations of materials and explainable ML concepts.

RESEARCH INTERESTS

- Computational Materials Science
- Data-Driven Discovery of Functional Material
- Scientific Machine Learning for Materials Design
- First-Principles Electronic Structure Methods
- Energy Storage Devices

SCHOLASTIC ACHIEVEMENTS AND AWARDS

- **Joint CSIR NET'2025:** AIR 63, Secured a Top 2 percentile (98.82) among 43,313 participants.
- **MCM Scholarship:** Awarded with Institute Merit-cum-Means Scholarship for being in the top 10% of students of M.Sc. (Chemistry).
- **IIT JAM'2024:** Secured AIR-198 among 14850 candidates across the country.
- **Recipient of the INSPIRE (Innovation in Science Pursuit for Inspired Research) Scholarship:** Awarded to the top 1% of students nationwide, given by Department of Science and Technology(DST), Government of India.
- **Medhavi Vidyarthi Samman(2019):** Awarded by the State Government of Uttar Pradesh for securing **Rank 8** at the district level (Prayagraj) in the Class X Board Examination.

RELEVANT COURSES

- Introduction to Computational Materials Science
- Quantum Chemistry
- Molecular Thermodynamics
- Solid State Chemistry
- Computational Techniques for Solid State Materials
- Computational Quantum Chemistry
- Cheminformatics
- Biophysical Chemistry

LEADERSHIP EXPERIENCE

- **Sponsorship Coordinator** May, 2025 - May, 2026
Chemocronies : Student Society of Chemistry Department, IIT Delhi 
- **Executive Mess Committee Member** May, 2025 - May, 2026
Zankar Hostel, IIT Delhi 
- **Executive** July, 2024 - May, 2025
Hindi Samiti, IIT Delhi 
- **Member of Organising Committee** November, 2024
Open House, IIT Delhi 

CONFERENCES AND WORKSHOPS

- **Attended ChemQuest** Annual symposium of the Department of Chemistry, IIT Delhi.
- **Attended International Symposium** on "Chemistry for Sustainability, Catalysis & Renewable Energy."
- Attended **Workshop on Advanced Theoretical & Experimental Research Tools (WATER2023)**, Department of Chemistry, University of Allahabad, 2023.

EXTRA-CURRICULAR ACTIVITIES

- *Strong communication and leadership skills* demonstrated through active participation in academic presentations, competitive debates, and organized sports activities.
- **2nd Position**, in Table Tennis at the Departmental Sports Tournament, IIT Delhi.
- Team Captain of the Zanskar Premier League (ZPL) cricket team; led the team to championship victory

REFERENCES

- **Dr. Dibyajyoti Ghosh**
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Relationship: [Masters Thesis Supervisor]
- **Dr. Sajesh P. Thomas**
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Relationship: [Course Instructor]